

PUNIT MISHRA

Senior Machine Learning Expert & AI Tech Lead — Agents, Inference & Multi-Tenant Platforms at Enterprise Scale

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PROFILE

Applied AI engineer and technical leader with 14+ years taking machine learning from prototype to dependable production at enterprise scale. Not only an architect — I **implement, deploy, and own systems end to end**, from agent runtimes and LLM inference pipelines through **GPU deployment on custom H100 clusters** and production scaling. As Senior ML Expert and AI Tech Lead in SAP CX Data & AI, I lead a team of thirteen *and personally build* the agent orchestration, memory, multi-model serving, and multi-tenant infrastructure beneath our LLM products — delivered in direct **forward-deployed partnerships**. Deep command of Rust, Kubernetes, distributed inference, vector search, and AI security.

EXPERIENCE

SAP	Sep 2018 – Present
<i>Senior Machine Learning Expert & AI Tech Lead — CX Data & AI (CXAI)</i>	<i>San Ramon, CA</i>

- **Built and own end to end** a domain-agnostic conversational agent platform — a memory-managed runtime with multi-tenant LangGraph orchestration, reusable across business domains, fronted by an **agent gateway and MCP-based service hub** that lets agents discover and compose backend capabilities behind one hardened contract; **implemented performance-critical paths in Rust**.
- **Introduced Ray** to the organization for distributed AI workloads and **built and deployed LLM serving on custom bare-metal H100 GPU clusters** — owning the full lifecycle from infrastructure bring-up through production inference, scaling, and a configurable embedding-router / self-hosted serving tier for latency- and cost-sensitive paths.
- Stood up **multi-model LLM serving** — self-hosted internal models alongside third-party frontier models (Claude, Qwen, Granite-class) on Kubernetes, with resolver-aware prompting and automated benchmark / evaluation harnesses across serving lanes.
- Shipped **natural-language-to-SQL** — turning business questions into validated queries — now shipping within SAP's **2026 public and private cloud releases**.
- Re-architected the platform's **vision capabilities** — visual-attribute extraction, object detection, and visual search — into first-class **gRPC agent services** exposed through the agent gateway and MCP hub; PyTorch and CLIP / text-image embeddings backed by **vector search and RAG** (Qdrant, FAISS).
- Designed the platform's **multi-tenant identity and provisioning**: a cross-service internal-auth contract and canonical tenant-resolution model unifying **IAS and XSUAA**, plus automated SaaS tenant provisioning (CAP / cds-mtxs sidecar) and a tenant registry.
- **Lead a team of thirteen engineers** across three workstreams; set architecture direction through ADRs, own backlog and code review across multiple GitHub orgs, mentor senior engineers, and **serve as forward-deployed engineer** on the platform's first production enterprise customer (a global retailer) — provisioning, security validation, and delivery.
- Established the **CI/CD and delivery backbone**: PR-test gates on ephemeral environments and **GitOps promotion** (dev → prod) via Argo and Piper on Kubernetes / Kyma (Gardener). **SAP Security certified**; own AI ethics & compliance and set the group's SDLC and engineering standards.

CALLIDUSCLOUD	2014 – 2018
<i>Senior Software Engineer & Architect — acquired by SAP for \$2.4B</i>	<i>SF Bay Area</i>

Built the Thunderbridge AI backend on Dataiku; designed multi-tenant ML and CI/CD across Oracle, Vertica, SAP HANA, and HDFS; ported the full ML stack to GKE on-prem (Anthos) with Istio.

IBM	2011 – 2014
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Built reporting and data-validation automation for JPMorgan Chase and reusable Selenium / Java test frameworks adopted across the firm.

SKILLS

AI & AGENTS	Agent orchestration (LangGraph, LangChain), MCP (agent gateway / tool hub), LLM serving & evaluation, NL-to-SQL, RAG, retriever-reranker (bge-m3), PyTorch, MLOps, Kubeflow
GPU & SERVING	Ray (distributed), custom bare-metal H100 clusters, Garden Linux, self-hosted LLM / embedding serving (Ollama), embedding-router tier, Knative / KServe
INFRA & CLOUD	Kubernetes, Kyma / Gardener, Istio, Argo / GitOps, Piper, Docker, Jenkins, CI/CD; vector DBs (Qdrant, FAISS), Redis; Azure (AKS), GCP (GKE / Anthos)
PLATFORM & DATA	gRPC / protobuf, CAP / XSUAA (OIDC / PKCE); PostgreSQL, SAP HANA, Oracle, Vertica, Databricks, Kafka, Polars
LANGUAGES	Python (FastAPI, Spark), Rust, Java, TypeScript / Next.js
SECURITY	SAP Security certified; internal-auth / identity design, threat modeling, SAST / SCA (GHAS / CodeQL, Coverity, Black Duck, Fortify)

EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY — B.S. Electrical Engineering & Computer Science	2012
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